The Monthly Average 7:00 a.m. Dew Point at Bryan, Texas: Two Years of Observation and a Twelve-Month Probabilistic Forecast

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Abstract

This report documents the monthly average 7:00 a.m. dew point at Bryan, Texas, over the two years ending August 2026, and presents a probabilistic forecast of that quantity for the twelve months from September 2026 through August 2027. The series is constructed from routine hourly observations at Coulter Field (KCFD), the automated weather station inside the Bryan city limits. The forecast comes from a seasonal harmonic model with a multiplicative seasonal variance term and a first-order autoregressive anomaly, simulated forward with bootstrapped innovations. The central result about the forecast is negative and worth stating plainly: at a monthly horizon this variable carries almost no exploitable persistence, so the forecast is close to a seasonal climatology with an honest, strongly season-dependent measure of dispersion. Its width is the substantive content. The 80 percent interval spans 3.9 °F in July 2027 and 15.2 °F in January 2027.[1]

1 Purpose and scope

The quantity of interest is the dew point temperature recorded at 7:00 a.m. local time at Bryan, Texas, averaged within each calendar month. The dew point is the temperature to which air must be cooled, at constant pressure and water vapor content, for saturation to occur; it is a direct measure of the absolute water vapor content of the air, unlike relative humidity, which confounds moisture with temperature. An early-morning reading is close to the daily minimum temperature, so it is the hour at which dew point and air temperature most often coincide and at which fog and dew formation are determined.

[1]This report was produced in response to the following prompt: "Create a report in this directory. Call this 'dew_point'. Use the 'soothfast' writing instructions in the https://github.com/h-bryant/latex-tools repo. The report should present a figure showing the evolution of the average monthly 7:00 AM dew point in Bryan, Texas, over the last two years, and a fan graph depicting a probabilistic forecast for that monthly value for the coming 12 months. Include this prompt in the a footnote." The prompt is reproduced verbatim, including its typographical errors.

Section 2 describes the observational record and the construction of the monthly series. Section 3 presents the two-year history. Sections 4 and 5 set out the forecasting model and the resulting predictive distributions. Section 6 evaluates the model out of sample, and Section 7 states its limitations.

2 Data

2.1 Station and series definition

The primary station is Coulter Field (ICAO KCFD, network identifier CFD), a municipal airport within the city of Bryan at 30.7157°N, 96.3314°W and 102 meters elevation. Observations are taken from the Iowa Environmental Mesonet archive of automated surface observations (Iowa Environmental Mesonet, 2026), restricted to routine hourly reports. Both stations report in METAR format through the federally sponsored automated airport observing networks, whose sensor suite, reporting cadence, and quality control are documented in the system user's guide (National Oceanic and Atmospheric Administration, 1998).

Coulter Field issues its routine observation at 55 minutes past the hour, so the 7:00 a.m. reading is defined here as the routine observation whose local valid time falls within fifteen minutes of 07:00, which in practice is the 06:55 report. Local clock time is used throughout, so the observation shifts by one hour in solar terms across the daylight saving transitions.

Two filters are applied to the raw readings. A dew point outside $[-20, 90]$ °F is treated as a sensor fault, as is any reading that exceeds the concurrent air temperature by more than the 0.1 °F reporting resolution. A calendar month enters the monthly series only if at least twenty days within it carry a valid reading.

2.2 Coverage and cross-station agreement

The Coulter Field record yields 4,746 daily 7:00 a.m. readings between 3 May 2013 and 30 August 2026, covering 97.5 percent of the days in that span, and 158 monthly means.

Coulter Field's record begins only in May 2013, which is short for estimating a seasonal climatology. A second station is therefore used as a longer reference: Easterwood Field (KCLL) in College Station, 14.5 kilometers south, whose archive runs from 1996 and yields 11,058 daily readings and 368 monthly means over the same construction.

The two stations agree closely. Over the 158 months both report, the correlation of their monthly means is 0.997, Coulter Field averages 0.29 °F moister, and the largest single-month discrepancy is 2.64 °F. That agreement is what licenses the use of Easterwood data in Section 4 to pin down two features of the Bryan series that its own record is too short to determine well.

3 The last two years

Figure 1 shows the monthly average 7:00 a.m. dew point at Coulter Field for the twenty-four months from September 2024 through August 2026, against the average for the same calendar month over the station's full 2013–2026 record.

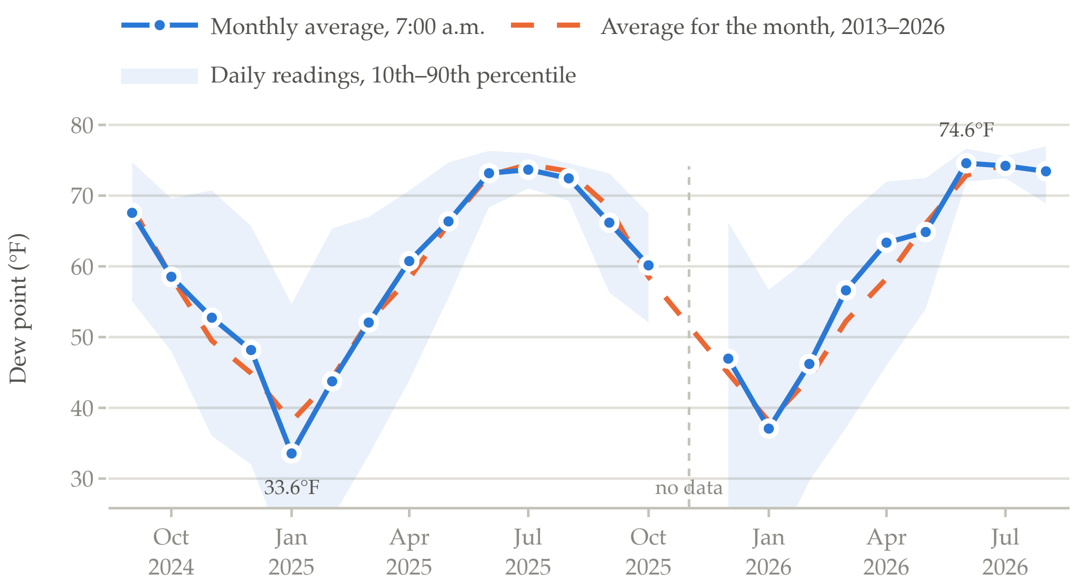

Figure 1: Monthly average 7:00 a.m. dew point at Coulter Field, Bryan, Texas, September 2024 through August 2026. The shaded band is the 10th to 90th percentile of the individual daily 7:00 a.m. readings within each month, not an uncertainty interval about the mean. November 2025 is absent because the station reported on only five days that month.

Twenty-three of the twenty-four months are observed. November 2025 is missing: Coulter Field returned a valid 7:00 a.m. reading on only five days that month, and those five days average 63.5 °F against 52.6 °F for the full month at Easterwood, so the surviving days are badly unrepresentative and the 20-day rule correctly excludes them. August 2026 rests on 30 of 31 days, the record having been downloaded before the month ended.

Two years is far too short a window to say anything about climate, and this report does not try to. What the window does show is an unremarkable pair of annual cycles. The mean over the twenty-three observed months is 59.4 °F, which is 0.6 °F above the station's own 2013–2026 average, and twelve of the twenty-three months sit above their calendar-month average. The extremes of the window are April 2026, 5.1 °F above its calendar-month average, and January 2025, 4.5 °F below. In levels, the window runs from 33.6 °F in January 2025 to 74.6 °F in June 2026.

The feature of Figure 1 that matters most for what follows is the shaded band. Within-month dispersion of the daily readings is small in summer and large in winter. In summer the morning air over central Texas is almost always maritime tropical air off the Gulf, and its moisture content varies little from day to day; in winter the same location alternates between that Gulf air and dry continental air behind frontal passages, and the daily readings scatter accordingly. The same asymmetry appears in the two-year record as a seasonal pattern in monthly mean variability, and any honest forecast interval must reproduce it.

4 A forecasting model for the monthly mean

4.1 Specification

Let y_t be the monthly average 7:00 a.m. dew point in month t, let $m(t)$ be the calendar month, and let τ_t be time in years. The model is

$$y_t = \mu_t + \sigma_t z_t, \tag{1}$$

$$\mu_t = \beta_0 + \delta\tau_t + \sum_{k=1}^{K}\left[a_k \cos\frac{2\pi k\, m(t)}{12} + b_k \sin\frac{2\pi k\, m(t)}{12}\right], \tag{2}$$

$$\log\sigma_t^2 = \gamma_0 + \sum_{l=1}^{L}\left[c_l \cos\frac{2\pi l\, m(t)}{12} + d_l \sin\frac{2\pi l\, m(t)}{12}\right], \tag{3}$$

$$z_t = \phi z_{t-1} + \varepsilon_t, \qquad \varepsilon_t \text{ independent, mean zero.} \tag{4}$$

Equation (2) is a Fourier representation of the annual cycle in the mean. Equation (3) is the multiplicative heteroskedasticity model of Harvey (1976), applied here to the calendar month rather than to a regressor; it is what encodes the seasonal asymmetry in dispersion described at the end of Section 3. Equation (4) allows a month-to-month carry-over in the standardized anomaly.

The numbers of harmonics K and L are chosen by the Bayesian information criterion. Both select a single harmonic. The mean equation is fit by ordinary least squares with heteroskedasticity- and autocorrelation-consistent standard errors (Newey and West, 1987), using six lags.

The scale equation's coefficients are estimated by regressing the log squared residuals on the harmonic basis and removing the -1.2704 bias that $\mathbb{E}[\log\chi_1^2] = \psi(1/2) + \log 2$ introduces.

4.2 Why the trend is not estimated from the Bryan record

The trend δ requires a decision. Estimated freely on the Coulter Field record, it comes out at $+1.61$ °F per decade with a HAC standard error of 0.71, apparently significant. It is not credible. Re-estimated as the record lengthens, the coefficient falls monotonically from

+7.25 °F per decade on data through August 2019 to +1.61 through August 2026 (Table 1). A genuine trend does not behave that way; a low-frequency swing being read as a slope does.

Table 1: The trend in the monthly mean 7:00 a.m. dew point, estimated on the Coulter Field record alone, as the sample lengthens. Degrees Fahrenheit per decade, HAC standard errors in parentheses.

Sample end	Months	Trend
August 2019	75	+7.25 (1.05)
August 2020	87	+6.54 (1.10)
August 2021	99	+4.51 (1.33)
August 2022	111	+3.74 (1.03)
August 2023	123	+2.70 (0.98)
August 2024	135	+2.64 (0.81)
August 2025	147	+1.98 (0.79)
August 2026	158	+1.61 (0.71)

The same specification fit to the 368-month Easterwood record gives +0.559 °F per decade with a HAC standard error of 0.242. That figure is of the order reported for United States surface dew point in the published literature, whereas the Coulter Field estimate is several times larger (Brown and DeGaetano, 2013; Gaffen and Ross, 1999).

The model therefore imposes δ at the Easterwood estimate rather than estimating it on the Bryan record, carrying the standard error of that estimate into the forecast simulation so the imposed trend is not treated as known. This is legitimate because the two stations are 14.5 kilometers apart and their monthly means correlate at 0.997: whatever the long-run slope at Easterwood is, it is very nearly the slope at Coulter Field.

The seasonal variance profile in equation (3) is borrowed from Easterwood on the same reasoning. Thirteen years supply only about thirteen observations per calendar month, and $\log \chi_1^2$ has variance $\pi^2/2$, so the shape of the seasonal variance profile is poorly determined at Coulter Field. Only the shape is carried over; the level is renormalized to Coulter Field's own residuals.

4.3 Estimates

The fitted annual cycle in the mean runs from 41.7 °F in January to 75.8 °F in July, evaluated at the end of the sample. The fitted standard deviation of the monthly mean runs from 1.57 °F in July to 6.51 °F in January, a ratio of just over four to one. The autoregressive coefficient is $\hat{\phi} = 0.217$ with a standard error of 0.078, so a month one standard deviation above its seasonal mean is expected to be about a fifth of a standard deviation above it the following month. A Ljung–Box test on the innovations at twelve lags returns $p = 0.054$, which is weak evidence of remaining structure rather than none.

The predictive distribution is obtained by simulation rather than from a normal formula. Each of 50,000 paths draws the mean-equation coefficients and ϕ from their estimated sampling distributions, draws the imposed trend from its own, and resamples ε from the empirical distribution of the fitted innovations, in the spirit of the bootstrap autoregressive prediction intervals of Thombs and Schucany (1990). Resampling rather than drawing normal shocks matters because the innovations are mildly right-skewed (0.23) and leptokurtic (excess kurtosis 0.86).

5 The twelve-month forecast

Figure 2 presents the forecast as a fan chart, the presentational convention introduced for the Bank of England's inflation projections (Britton et al., 1998) and since standard for density forecasts (Wallis, 1999; Tay and Wallis, 2000). Table 2 gives the same information numerically.

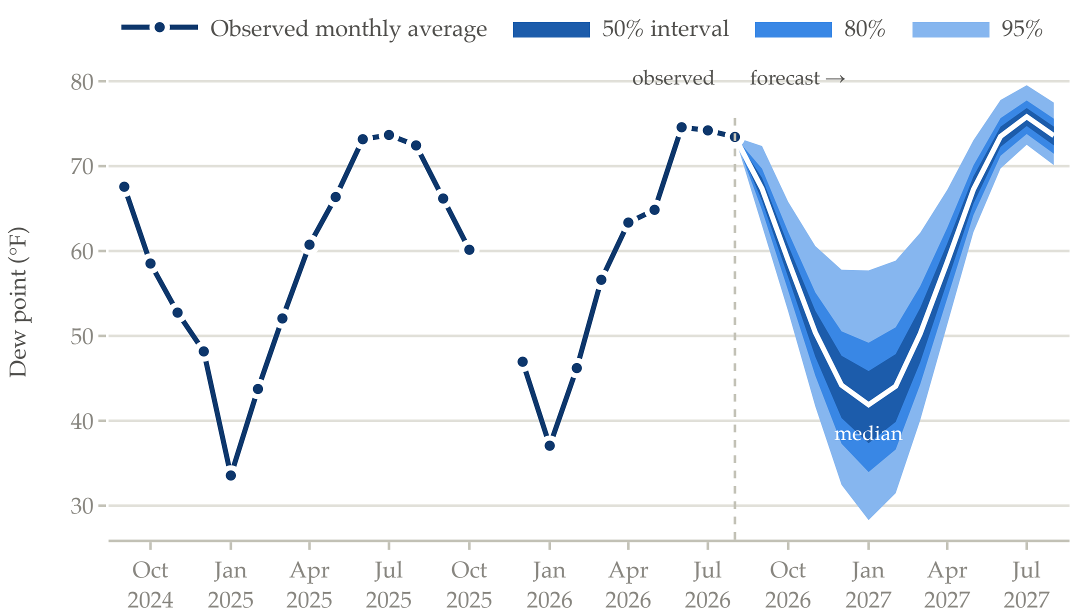

Figure 2: Fan chart of the monthly average 7:00 a.m. dew point at Coulter Field, Bryan, Texas. Observed values through August 2026; simulated predictive distribution for September 2026 through August 2027. Bands are the central 50, 80, and 95 percent of the predictive distribution, shaded darker where the outcome is more probable. The fan is anchored at the last observed month, whose value is known.

The medians trace the seasonal cycle almost exactly, and this is the honest result rather

Table 2: Predictive distribution of the monthly average 7:00 a.m. dew point at Coulter Field, Bryan, Texas, in degrees Fahrenheit. Central intervals from 50,000 simulated paths.

Month	Median	50% interval	80% interval	95% interval
September 2026	67.3	65.9–68.6	64.8–69.7	63.1–72.4
October 2026	58.9	56.9–60.7	55.3–62.2	52.9–65.8
November 2026	50.4	47.5–53.0	45.3–55.1	41.7–60.6
December 2026	44.2	40.3–47.7	37.3–50.5	32.4–57.8
January 2027	41.9	37.4–45.9	34.0–49.2	28.3–57.7
February 2027	44.1	39.9–47.9	36.6–51.0	31.5–58.9
March 2027	50.2	46.9–53.3	44.3–55.8	40.0–62.1
April 2027	58.7	56.3–60.9	54.4–62.7	51.5–67.2
May 2027	67.2	65.6–68.8	64.2–70.1	62.2–73.1
June 2027	73.5	72.3–74.6	71.2–75.6	69.7–77.8
July 2027	75.8	74.7–76.8	73.8–77.7	72.5–79.5
August 2027	73.6	72.4–74.7	71.5–75.6	70.1–77.5

than a failure of effort. The standardized anomaly in the final observed month, August 2026, is -0.06, essentially zero, so even the first-month forecast has nothing to carry forward; and with $\hat{\phi} = 0.217$, any carry-over would have decayed to under five percent of its initial size within three months in any case.

The information in Figure 2 is therefore almost entirely in the width of the fan, which varies by a factor of four across the year. The 80 percent interval spans 3.9 °F for July 2027 and 15.2 °F for January 2027. A user planning around next July's morning humidity can treat 75 °F as close to a sure thing; a user planning around next January's cannot usefully treat any value between the middle thirties and the high forties as unlikely. The fan is also visibly right-skewed in winter, its 95 percent upper bound sitting further from the median than its lower bound, because the log-variance specification combined with right-skewed innovations produces asymmetric intervals in the high-variance months.

6 Out-of-sample evaluation

The model was evaluated by expanding-window pseudo-out-of-sample forecasting. At each origin from August 2019 through July 2026, every quantity is re-estimated from data dated on or before that origin, including the imposed trend and the borrowed variance profile, and forecasts are issued for horizons one through twelve months. That yields 930 forecast-target pairs. Specifications are ranked by the continuous ranked probability score, which scores the whole predictive distribution rather than a point forecast (Hersbach, 2000; Gneiting and Raftery, 2007), and checked with the coverage of the central intervals and the probability integral transform (Diebold et al., 1998).

Table 3 reports the comparison for the specifications that matter. One difference dominates. Extrapolating the freely estimated trend raises CRPS by 0.38, or 21 percent, and leaves a mean forecast error of +1.67 °F: the model systematically forecasts too moist, exactly as Table 1 predicts it would. Every specification that declines to extrapolate that trend performs within noise of every other, across a band from 1.784 to 1.856.

Table 3: Pseudo-out-of-sample comparison, 930 forecast-target pairs, origins August 2019 through July 2026. CRPS and errors in degrees Fahrenheit; nominal interval coverage is 0.50, 0.80, and 0.95. The selected specification is the first row.

Trend	Variance profile	CRPS	MAE	Bias	Cov. 80%	Cov. 95%
Anchored	Easterwood	1.784	2.504	−0.26	0.898	0.970
None	Easterwood	1.796	2.520	−0.44	0.900	0.975
Anchored	Coulter	1.833	2.567	−0.14	0.902	0.973
None	Coulter	1.844	2.591	−0.34	0.899	0.971
Free	Coulter	2.167	3.008	+1.67	0.697	0.926

Three further results deserve statement.

First, the autoregressive term earns nothing. Pooled across horizons the model scores 1.783 with it and 1.775 without; at a one-month horizon it scores 1.887 with and 1.843 without. The in-sample $\hat{\phi} = 0.217$ is real but too small to survive estimation error out of sample. The term is retained because it is the pre-specified structure and because removing it on the strength of a difference this small, measured on heavily overlapping pairs, would itself be a form of overfitting; but no claim is made that it improves the forecast.

Second, the intervals are somewhat too wide rather than too narrow. Pooled coverage of the 50, 80, and 95 percent intervals is 0.568, 0.896, and 0.968 against nominal 0.50, 0.80, and 0.95. Restricting to origins from August 2023 onward, where the training sample is closer in length to the one actually used, coverage is 0.497, 0.867, and 1.000 on 354 pairs. Erring wide is the conservative direction for a fan chart, but readers should understand that the bands in Figure 2 are, if anything, a little generous.

Third, accuracy does not degrade with horizon at all. Mean absolute error is 2.70 °F at one month and 2.44 °F at twelve, and does not trend in between. That is the same finding as the first, seen from another angle: nothing in this month's observation tells you much about next month's, so a twelve-month-ahead forecast is no harder than a one-month one.

7 Limitations

The seasonal mean uses a single harmonic, which BIC selects and which the out-of-sample comparison does not reject, but which cannot reproduce the sharper minimum the data show in January. In the backtest the model runs +2.99 °F too moist in January and −3.05

and -2.86 °F too dry in March and December, against a mean error below 1.2 °F in absolute value in every month from April through November. Adding a second harmonic reduces those winter biases but does not improve the overall score, because the winter months where it helps are also the months with the widest intervals, where a mean shift of two or three degrees costs little. Users interested specifically in midwinter should treat the December through March medians in Table 2 as carrying a systematic error of a few degrees whose sign is known.

The forecast conditions on no information beyond the station's own history. It uses no sea surface temperature state, no El Niño–Southern Oscillation index, and no numerical weather prediction or seasonal model output, any of which would plausibly narrow the winter intervals where the uncertainty actually lies.

The record is short and single-sited. Thirteen years at Coulter Field cannot distinguish a trend from a low-frequency swing, which is precisely the problem Section 4 works around rather than solves. Nor has any check been made for instrumentation changes, site changes, or calibration drift at either station, any of which would show up as an apparent trend.

Finally, the specification was selected using the same backtest that then reports its performance. The 930 pairs overlap heavily, so the apparent precision of the comparison in Table 3 is overstated, and the reported scores for the selected model are optimistic by an unquantified amount.

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